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GNSS error simulator for farm machinery navigation development

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ABSTRACT

Satellite positioning systems are widely used for determining the position of moving machines and vehicles. Different methods for improving the accuracy and reliability are often used by fusing dead reckoning systems to the satellite positioning. To develop these fusions and to determine sufficient positioning accuracies, a realistic simulator for positioning data is required. In this study, we developed a GNSS (Global Navigation Satellite System) error simulator to support the development of navigation capabilities. The simulator was based on real dynamic tests driven with an agricultural tractor. The driving speed of the tests was approximately 9 km/h and the GNSS measurement rate was 10 Hz. We used Allan variance calculations to determine the characteristics of the errors and Geary's C to determine the spatial autocorrelation of the error in agricultural fields. With these methods, we constructed a feasible error simulator capable of simulating positioning errors and positioning reliability for all typical GNSS positioning accuracy levels. The simulator uses White noise, Random walk, Flicker noise, Rate ramp, a First order Gauss–Markov process and the spatial autocorrelation map to generate errors for horizontal coordinates.

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1. Introduction

Global Navigation Satellite Systems (GNSS) are widely used for the position estimation of outdoor machines such as agricultural machinery and vehicles such as mobile robots, marine vehicles and aerial vehicles. The positioning quality depends on the available GNSS satellite constellation and its variation, and the quality of signals from different satellites. Single signals include errors caused by atmospheric effects, multipath effects, ephemeris errors and clock errors. For many purposes, the accuracy and reliability of autonomous GNSS positioning is not enough. The exploited correction systems of the GNSS positioning and possible supporting measurement systems vary a lot. The selection is often a balance between the expenses of the system and the expected accuracy and efficiency levels of the system.

In machinery navigation purposes, the entire machine or vehicle moves from location to another in varying GNSS satellite visibility conditions usually by following a pre-planned route. The reasonable applications are steering assistance, automatic steering, semi-autonomous machinery and robotics. To manage with errors and uncertainties in GNSS positioning, it is common to combine relative positioning methods (Mousazadeh, 2013). Typically dead reckoning methods (Seung et al., 2013) are fused to the GNSS positioning. The fusion is usually made by applying different types of Kalman filters. The standard Kalman filter requires the fused measurement noises to be white noise. GNSS errors are often represented as white noise (Groves, 2008). However, a simple white noise is not always sufficient (Oksanen et al., 2005 and Niu et al., 2013) since almost all of the GNSS positioning error sources are time or space correlated (Rankin, 1994 and Bierman, 1995). It would be essential to have a realistic GNSS error simulation to be able to develop operational sensor fusions.

Beside typical GNSS accuracy tests, machinery mobility and the changing environment set challenges for the GNSS error definition. Min et al. (2008) made dynamic GPS tests with a tractor in citrus orchards and found that receivers performed differently under various orchard conditions. Pirti (2008) found that tree canopy on one side increased the standard deviation of static position around 40%. The work of Stombaugh et al. (2002) showed that a static performance of receivers might not be even indicative of dynamic performance and there could be a need of a dynamic GNSS test standard.

Abbreviations: ADEV, Allan standard deviation; DOP, dilution of precision; FN, Flicker noise; GM, first order Gauss Markov process; GNSS, Global Navigation Satellite System; HP, high performance; MBR, minimum bounding rectangle; NMEA, National Marine Electronics Association; OL, outliers; PSD, power spectral density; RR, Rate ramp; RTK, real time kinematics; RW, random walk; VBS, virtual base station; WN, white noise.

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To determine the real GPS error used on a mobile machine, Bell (2000) evaluated the control system's accuracy by the mean and standard deviation of the GPS measured tracking error from the desired trajectory. Since that, there have been various dynamic tests for different steering systems of the tractor (Kaivosoja, 2008) and currently there are standards for agricultural machinery, such as the dynamic testing of satellite-based positioning devices (ISO, 2010), and the testing of satellite-based auto-guidance systems during straight and level travel (ISO, 2012). In general, these test standards would be a suitable basis for gathering data for the GNSS error simulation. However, those tests are meant for benchmarking different receiver units aiming to give comparable information based on a defined test environment.

To be able to exploit the determined GNSS errors, a methodology for the error generation is needed. Rankin (1994) constructed simulations which modeled GPS error statistics for various receivers. The simulator had a model of GPS satellite orbits, which were used to create the dilution of precision (DOP) values that translated pseudo range errors to X–Y–Z errors which the simulator generated. The works of Oksanen et al. (2005) and Backman et al. (2010) introduced GPS noise simulations for agricultural tractors. In those studies, different state-space models were fitted to approximate real measured errors as well as possible. Backman et al. (2010) simulated also the number of visible satellites, which had an effect on the temporal error. Roberts et al. (2002), Niu et al. (2010), Khelifa et al. (2011) and Niu et al. (2013) showed that Allan variance is a feasible way to analyze the static GNSS positioning characteristics. Based on Allan variance interpretations, Niu et al. (2013) identified four dominant noise terms from the GNSS positioning error: a First order Gauss–Markov process, Gaussian white noise, random walk noise and Flicker noise. It would be beneficial to adapt Allan variance analyses to the error simulation in a dynamic environment of mobile machines.

In addition, taking into account the local variations of error would be essential. This variation can be caused by the surrounding forests, buildings, hills, signal multipath sources, slopes in the field and different obstacles inside the field parcel. A spatial autocorrelation represents a degree of dependency among observations on a geographic space. Close objects are similar to each other. In positioning errors in machinery navigation, this means that parallel driving tracks in certain areas would have similar amounts of error. The level of spatial autocorrelation can be indicative of the local error behavior.

It would be beneficial to construct a realistic simulator which could simulate different correlation and reliability levels changing conditions. The research questions are as follows: (1) what are the Allan variance elements for different dynamic GNSS positioning methods, (2) how to replicate the found Allan variances, (3) what is the spatial autocorrelation effect on the amount of local error, (4) what is the structure of the error simulator capable of handling determined Allan variances and spatial autocorrelation being also complex to retrace?

This study introduces a new tool for supporting the development of sensor fusions in steering and navigation automation by implementing Allan variance methodology and spatial autocorrelation determination to reveal positioning error characteristics based on dynamic tests in farming conditions. The presented simulator is capable of simulating autonomous GNSS positioning in bad conditions and in normal field conditions, sub-meter level differential (DGPS) positioning and nominally high accurate RTK (real time kinematics) GNSS positioning in various conditions. Free SBAS's (EGNOS) were not used in this simulator because of the challenging connection stability in Finland. Our study differs from previous research by implementing the Allan variance interpretation into the dynamic environment and by adapting the spatial autocorrelation to the error formulation.

2. Materials and methods

First we introduce methodologies for analyzing the characteristics of positioning errors. Then we present our data collection methods and gathered data. After that, we calculate the Allan variances of errors and determine the dominant noise terms found in different tests. Then we reconstruct the Allan variances by adjusting the found noise terms. After that we measure the amount of spatial autocorrelation in four test drives and reconstruct the effect of it. Finally, we design and construct an adjustable error simulator. Next we present these materials and methods in detail.

2.1. Methodology for analyzing errors

The Allan variance is a two-sample variance (Lesage and Audoin, 1979) and is a time-based domain analysis technique originally developed to study the frequency stability of precision oscillators. The Allan variance analysis has been widely accepted as a method of identifying stochastic processes (Niu et al., 2013). A detailed definition and explanation of the Allan variance are presented in IEEE (2008). The Allan variance presents how much there is variation in average between clusters with certain length. The equation for the estimation of Allan variance with the fixed τ estimator is:

$$\hat{\sigma}_x^2(\tau, M) = \frac{1}{2(M-1)} \sum_{i=0}^{M-2} (\bar{x}_{i+1} - \bar{x}_i)^2,$$

where M is the number of averaged clusters in data and $\bar{x}_k$ is the mean value of the clusters. To analyze the error characteristics, a log–log plot of the Allan deviation is used (Niu et al., 2013). The plot is called Allan variance plot. Allan deviation (ADEV) is a square root of the Allan variance.

The examination of the Allan variance plot allows different noise components to be distinguished by the slope of the plot in particular time regions and in different magnitudes (Land et al., 2007). Gaussian white noise is characterized by a region of $-1/2$ slope, Random walk noise is associated with +1/2 slope and Flicker noise is shown as a region of 0 slope (IEEE, 2008 and El-Sheimy et al., 2008). The frequency spectrum of Flicker noise is linear in logarithmic space. The fourth found type of noise presented by Niu et al. (2013) was a First order Gauss–Markov (GM) process, which appears as raised points on the Allan variance plot (Niu et al., 2013). The GM processes are stochastic processes that follow both Gaussian and Markov processes. GM processes have an exponential autocorrelation function with variance and correlation time (Rankin, 1994). The equation for GM process is:

$$R(\tau) = \sigma^2 e^{-\beta|\tau|},$$

where σ^2 is the variance and the Gauss–Markov terms are modeled as:

$$x_{k+1} = e^{-\beta+\Delta T} x_k w_k$$

where x_k is the simulated error, w_k is Gaussian white noise, ΔT is the sample time and $1/\beta$ is the correlation time that will be determined for the simulations together with the standard deviation σ. The correlation time has typically been around 600–3600 s with static GPS positioning (Cannon and LaChapelle, 1992).

Borre and Tiberius (2000) have stated that GPS receiver's noise at sampling rate below 1 Hz is independent and does not have any time correlation. The most possible reason for a time-correlated noise is the remaining tropospheric path delay (Howind et al., 1999). According to Rankin (1994) that can be modeled as the First GM process.

Although the Allan variance analysis can give great estimations in general noise determination, El-Sheimy et al. (2008) stated that

Allan variance does not always determine a unique noise spectrum since the spectral characterization is limited. This put restrictions on what can be learned about a noise from the examination of its Allan variance. However, Niu et al. (2013) concluded that the Allan variance analysis method can be proposed for the GNSS positioning, with some conventional method, for example, PSD (power spectral density) as supplement if necessary. The PSD describes how the squared value of a noise is distributed directly over the different frequencies. It shows at which frequencies variations are strong and which are weak. Niu et al. (2013) showed that the static GNSS noise types identified by the PSD and the Allan variance methods were the same. Based on this, we used PSD calculations only for the evaluation of our results. We calculated plots for each simulated and measured data to evaluate a general similarity of the spectrums in this study. In this case, the PSD is a square of the absolute values of Fourier Transform over a large time interval (T). The PSD formula for the $x(t)$ signal can be presented as:

$$S_x(f) = \lim_{T\to\infty} E\left\{\frac{1}{2T}\left|\int_{-T}^{T} x(t)e^{-j2\pi ft}dt\right|^2\right\}.$$

On top of these presented noise characteristics based on static positioning, we detected two types of noises from the dynamic tests. We detected spatial dependencies from each test data and separate noise peaks in our most challenging test conditions. The peaks were simulated separately in straight forward way, but the spatial dependencies required methodology development.

The classic spatial autocorrelation statistics compare the spatial weights with the covariance relationship at pairs of locations. There are two main approaches: a raw-data approach and a matrix approach (Legendre, 1993). For the correlation determination, we used the raw data approach where the spatial structure takes the form of a free X and Y geographical coordinates of the samples. To exploit the correlation, we used the matrix approach, where the spatial structure is in the form of a geographic distance matrix among locations.

As a typical autocorrelation determination, we calculated both Geary's C (Geary, 1954) and a Moran's I (Moran, 1950) for all of those four datasets. Those are inversely related, but Geary's C is more sensitive to local spatial autocorrelation and we estimated it to better indicate the spatial autocorrelation of error in mobile positioning. There are also other methods for estimating the autocorrelation such as Local Moran's I and Getis-Ord Gi* but we expected Geary's C to reveal autocorrelation at sufficient level in relation to our limited amount of data. The equation of Geary's C is:

$$C = \frac{(N-1)\sum_i\sum_j \omega_{ij}(Z_i - Z_j)^2}{2W\sum_i (Z_i - \overline{Z})^2},$$

where N is the number of measured points indexed by i and j and Z is the amount of error. Variable $\overline{Z}$ is the mean error and ω_{ij} is a matrix of spatial weights. In this study, all observations within a specified distance (*Lag* in Table 4) have a fixed weight 1. W is the sum of all of those spatial weights.

The result of Geary's C is between 0 and 2. Values lower than 1 demonstrate increasing positive spatial autocorrelation, whilst values higher than 1 illustrate increasing negative spatial autocorrelation.

2.2. Data collection

We collected both static and dynamic GNSS positioning error data. As dynamic tests, we used data from our tractor test drives from past eight years. In each test, we recorded NMEA (National Marine Electronics Association)-type GNSS-messages. In our principal dynamic test setup, two high accurate RTK (real time kinematics) GNSS antennas were located on both sides of the tractor's roof. The test unit was located in the middle (Fig. 1). The true position of the test unit was determined based on the positions and orientation of the two RTK receivers. The determined error was the difference between the measured true position and the recorded position of the test unit at the same time.

Data collection and synchronization were executed in National Instrument LabVIEW environment used on a Cropinfra platform (Pesonen et al., 2013). The test drives were close to regular farming tasks, following the GNSS test standard ISO/DIS 12188-1 by the driving speed and by the accuracy of the reference system, which was more than ten times better. We selected the driving speed for test drives to be approximately 9 km/h since it is relatively common and safe for agricultural tractors and equals the middle test speed in the dynamic test standard (ISO, 2010).

For data collection, we used seven GPS receivers in total (Table 1). As reference position measurements, we used two Trimble 5700 receivers together with a Trimble SPS852 receiver. These units used RTK–VRS (Virtual reference station) correction with centimeter level accuracy. We used those receivers also to measure correction drop-offs and to make static RTK tests. The supporting GLONASS positioning in Trimble SPS852 was always used. Two Trimble 7400 receivers with a local RTK-reference station were used with our first tests in 2006.

As the main test unit for dynamic tests, we used a low-cost Garmin 19x receiver in the autonomous GPS mode. The second test unit was NovAtel receiver with a decimeter level OmniSTAR high performance (*HP*) differential GPS correction being quite typical for the guidance usage in agriculture. Also one test with OmniSTAR sub-meter level *VBS* (virtual base station) – correction was used.

The limiting elevation masks of the receivers were selected to be 10 degrees and the DOP values were not limited. The test setups are presented in Table 1 including the name of the test setup, the test unit, the positioning mode, measurement intervals (*Rate*), test area, the reference positioning system and the year when data was acquired.

Aut and *AutE* were autonomous GPS measurements using L1 frequency, the *VBS* method uses a satellite based augmentation system (SBAS) as the correction source and L1 frequency only, while the *HP* method used SBAS and both L1 and L2 frequencies, the *RTK* in Fixed mode positioning calculates the exact number of wavelengths between the satellites and the receiver. In *RTK_Float* measurements, the poor positioning conditions prevented the fixed solution leaving ambiguity to the solution giving only a Float point number.

All the dynamic positioning tests representing results in typical positioning conditions were performed with Garmin 19x and Omnistar HP receivers (*Aut* and *HP* setup in Table 1) in LUKE (former MTT Vakola) by repeating a route on a test track.

In *AutE* test (Table 1), we performed an extreme positioning performance test. The driving path was circulating buildings also by temporally going inside to two machine halls.

The RTK-level performance (*RTK* and *RTK-Float* in Table 1) were carried out as static measurements, since there were no better reference measurement systems for them. Tests *HP2006* and *VBS2006* (Table 1) were performed on an agricultural field and were used for spatial autocorrelation determinations.

Table 2 presents the number of measurement points in each test and the found mean and standard deviation of the error for latitude and longitude measurements in meters. The longitude coordinates were less accurate than the latitude coordinates on average.

To determine the positioning mode reliability of the test unit and reference units, we analyzed the *AutE* test data. To support this, we performed a 3-h static test with the same instrument setup and an eight-hour static test with the RTK–VRS receiver SPS852 where great parts of the sky were blocked.

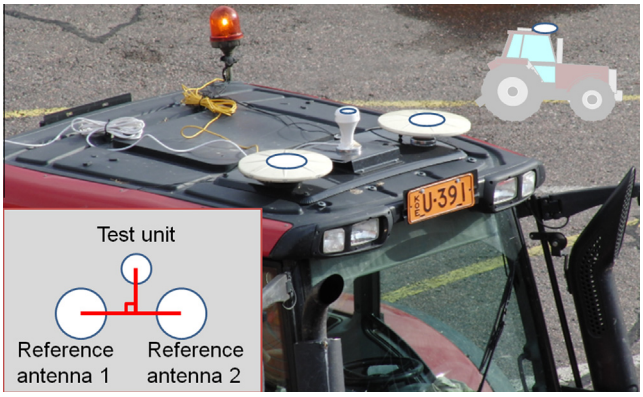


Fig. 1. Measurement setup showing the position of GNSS antennas.

Table 1
Test setups of applied test drives.

Name	Tested GNSS	Mode	Rate	Area	Reference	Year
Aut	19x	Autonomous	10 Hz	Track	5700 + SBS852	2013
AutE	19x	Autonomous	10 Hz	Yard	5700 + SBS852	2013
VBS2006	NovAtell	VBS	3 s	Field	2∗7400	2006
HP	NovAtell	HP	10 Hz	Track	2∗5700	2010
HP2006	NovAtell	HP	3 s	Field	2∗7400	2006
RTK	5700, SBS852	VRS	10 Hz	Static	1∗SBS852	2013
RTK_Float	5700	aut./Float/Fix	10 Hz	Static	1∗SBS852	2013

The positioning mode reliability results from these tests are presented in Table 3 showing how long in percentage the receiver managed to keep a fixed solution and how much time it was not providing positioning data at all. Although giving positions all the time, 19x receiver was drifting a lot.

In the 8-h test, the longest constant period of fixed solution was 57 min. The longest constant period of Float mode was 84 s. There were 35 drops to Float mode lasting about 520 s in total. The maximum leap in latitude was 85 cm and in longitude 30 cm. The median leap was 2.8 mm.

We also made static and dynamic reliability tests with Egnos and HP. The Egnos correction was basically not exploitable at all in our test area. After successful establishment, the HP-correction was always available, even when only four satellites were in use.

2.3. Allan variances

Next we calculated the Allan standard deviation plots for the presented test drives. The plots are presented in Fig. 2. It can be seen from the plots that Allan variances are behaving very differently with different test data sets and with different correction methods.

Based on calculated Allan Variances, we calculated the slopes between different cluster times. The slope values are determined as $\Delta\log(y)/\Delta\log(x)$ of calculated Allan deviations for each cluster time. Table 4 shows the calculated Allan variance slopes and shows the closest noise centers that we manually pointed out by examining both absolute values and separate Allan variance plots that revealed the overall characteristics.

Table 3
Positioning time in different modes (percentage).

Receiver	Test type	RTK fix	RTK float	Autonomous	No data
SPS852	Dynamic	60	22	16	1.9
T5700	Dynamic	68	26	3.2	3.2
19x	Dynamic	–	–	100	–
SPS852	Static	94	4.5	1.1	–
T5700	Static	98	0.7	1.3	0.01
19x	Static	–	–	100	–

In addition to noise terms (white noise −0.5, flicker noise 0, random walk 0.5) discovered by Niu et al. (2013) in static tests, the RTK-Float measurements had also close to a +1 slope, which can be interpreted as Rate ramp. We also found some outliers from the coarsest data. We processed those outliers separately by determining the number of big leaps on data and the power and duration of them. We removed the detected outliers from source data and simulated them separately according to found characteristics.

The impreciseness of the exact slope values may be caused by adjacent noises being dominant in longer cluster times (Niu et al., 2013). Also weaker noise terms may namely disappear due to multiple other noise terms or complicate other noise terms.

2.4. Error reconstruction

Next we simulated the errors in each dynamic test drive as closely as possible. We used the found dominant noise types (Table 4) as the starting values for the simulation constructions. From test data, we identified six possible different noise types:

1. WN, Gaussian distributed white noise
2. RW, Gaussian distributed random walk steps
3. FN, Flicker noise, Random noise which frequency spectrum is linear in logarithmic space
4. GM, First order Gauss–Markov process
5. RR, Rate ramp, linear drifting
6. OL, Outliers, large peaks in data

One simulation can contain multiple same noise types with different parameters and random seeds. We limited the total number of noise sources for a single simulation to be five. We estimated that to be enough to construct a complex simulation for retracing. Niu et al. (2013) sufficiently simulated static measurements with three noise sources.

We tuned the parameters by selecting two cost functions for the optimization: (1) the standard deviation of differences in percentage between real and simulated data in separate cluster times 1–128, (2) the average of the differences. The maximum accepted value for both of them were 10% calculated from the real values. To be able to have an estimation of relevant variation we split each data in half and compared the Allan variances of first and last half of data.

The cost functions were minimized with Matlab Simulink by manually tuning the parameters. First, we selected three to four main noise terms based on determined noise terms in Table 4 and gains in Fig. 2, then we tuned the parameters to minimize the cost functions and to match the simulation to the original

Table 2
Mean error and standard deviation of the error in the test drives (in meters).

Name	AutE_lat	AutE_lon	Aut_lat	Aut_lon	HP_lat	HP_lon	Float_lat	Float_lon	RTK_lat	RTK_lon
Points	5839	5839	14,126	14,126	18,008	18,008	824	824	20,655	20,655
Mean	0.417	1.543	0.598	1.528	0.101	0.309	0.065	0.034	0.007	0.004
SDV	0.452	0.451	0.217	0.377	0.123	0.113	0.055	0.028	0.009	0.005

Table 4
Measured Allan variance slopes and found centers of the noise terms in bold.

t	AutE_lon	Aut_lon	HP_lon	Float_lon	RTK_lon	AutE_lat	Aut_lat	HP_lat	Float_lat	RTK_lat
1–2	0.18	**0.07**	−0.24	−0.20	**−0.53**	0.18	0.17	**−0.54**	−0.11	**−0.55**
2–4	0.33	0.20	**−0.51**	0.19	−0.10	0.26	0.30	−0.74	0.18	−0.15
4–8	**0.43**	0.35	−0.53	**0.34**	0.26	0.32	0.43	−0.27	0.29	0.21
8–16	0.41	**0.52**	0.15	0.66	0.18	0.31	0.53	**−0.03**	**0.50**	0.21
16–32	0.26	0.54	**0.48**	0.87	0.24	0.30	**0.51**	0.16	0.82	0.32
32–64	0.21	0.45	0.34	**0.94**	0.28	**0.43**	0.43	**0.37**	0.95	0.29
64–128	**−0.09**	0.43	0.22	0.87	0.10	0.38	0.23	0.36	**0.98**	0.18
128–256	−0.16	**0.46**	0.14	0.24	**0.05**	0.18	**0.08**	0.27	1.10	0.25
256–512	0.20	0.43	0.42		0.13	−0.80	−0.06	0.33		0.27
512–1024	0.22	−0.17	0.55		0.39	**−0.46**	**0.01**	**−0.03**		**0.45**
1024–2048		**−0.02**	**0.50**		**0.47**		−0.95	−0.21		**0.09**
2048–4096		−1.88	−0.13		−0.23		−0.25	**0.09**		**−0.93**
Local high	70, 1000	500, 1900	2100	–	1000	150	250, 1000	600, 2500	–	2000

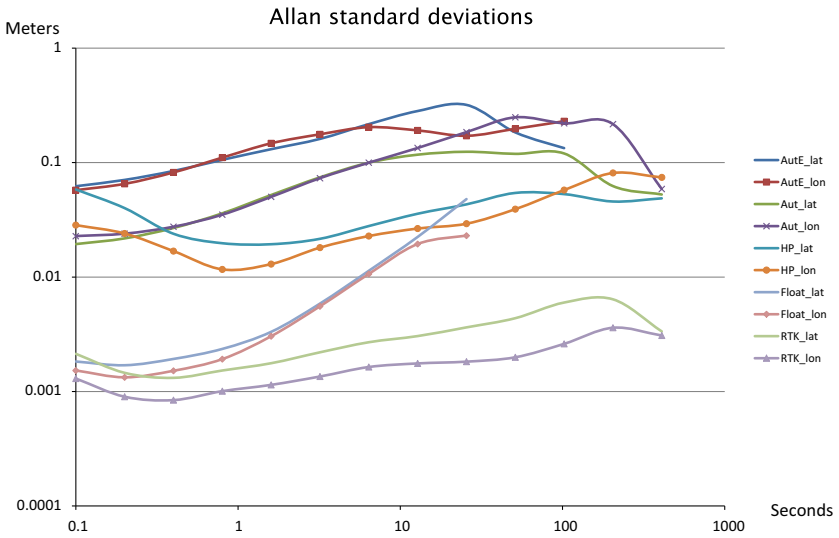


Fig. 2. Allan standard deviation plots of the test drives.

graph. This manual tuning was based on the concurrent evaluation of the generated Allan variance curve, the clearness of the noise centers in Table 4 and the test drive plot curves (Fig. 2). Less clear noise centers were adjusted first (power and/or frequency) and the effect was then evaluated.

Fig. 3 presents four consecutive optimization steps for *Aut_lat*. First, the update rate of other Gauss Markov simulation was increased to increase the noise power in the middle cluster times and then the characteristics of Random Walk noise were changed by nearly doubling the standard deviation to still increase the slope, but also adding a reset time to 250 to minimize the effect in greater cluster times. As the last tuning step, nearly all of the noise terms were adjusted slightly meaning there were practically many small tuning steps. Also finally a rather weak new White noise was introduced to equally increase the noise levels in different cluster times. This manual tuning was mandatory because of the unknown interaction between different noises in different cases.

The roles and behavior of different noise terms for the *AutE_lon* simulation are illustrated in Fig. 4. This shows that each noise term has an influence on each cluster time. The used parameters are shown later on a results section (Table 6).

We used Gaussian smoothing for WN, RW and FN noise terms that had a detected frequency greater than ten measurements. To ensure the difficulty for retracing, we made small frequency variations for some of the noise terms by randomizing the new value entry time. This softened the noise effect in specific frequencies.

2.5. Spatial autocorrelation calculations

For the spatial autocorrelation determination, we analyzed four different dynamic tests. Two of them were driven on a field (*HP2006* and *VBS2006* in Table 5), and two other were on a test track (*HP* and *Aut* in Table 5) driving along the same track for six to eight times respectively.

First, data were transferred to metric coordinates. Then total amount of error in meters for each point was calculated. The number of measurement points on field test data was reduced by calculating a mean error value for every third measurement to approximately match the distance between consecutive and parallel points being 20 m in average (Fig. 5). The positioning error in Fig. 5 varied between 1.0 and 2.8 m having local variations.

The presented *Lag* in Table 5 represents the maximum Euclidean distance between two cells. Next we constructed a thematic map approximating the autocorrelation levels sufficiently. We used a method called an unconditional Gaussian simulation to construct a spatially correlated random field named *A-Map* (Fig. 6) by using *R* software. Demonstrative studies of unconditional Gaussian simulations with *R* are presented by Begueria (2010) and the mathematics behind are published by Schlather (2012). For the *A-Map*, We selected a grid size of 25×25 to match the resolution we measured the real spatial autocorrelation to be able to repeat the spatial autocorrelation characteristics, which depends on data collection resolution and the work extent (varying canopy around the field). We used an exponential variogram model with parameters:

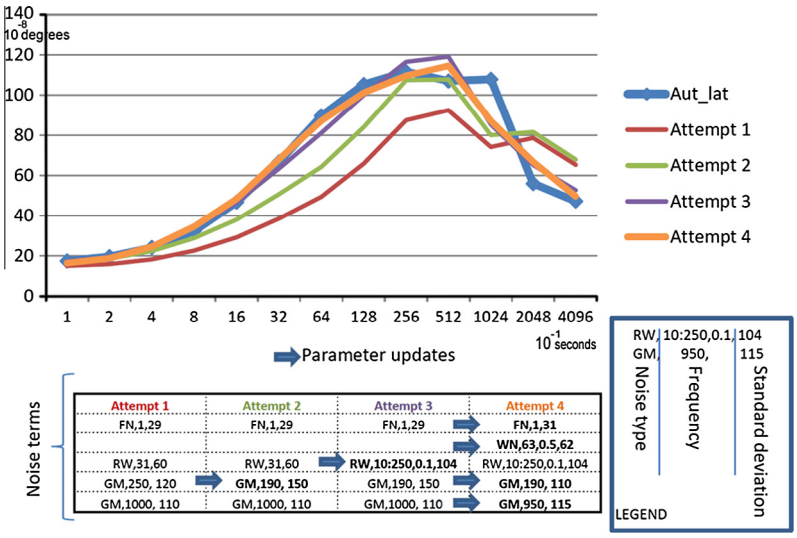


Fig. 3. Three tuning steps in the error reconstruction process.

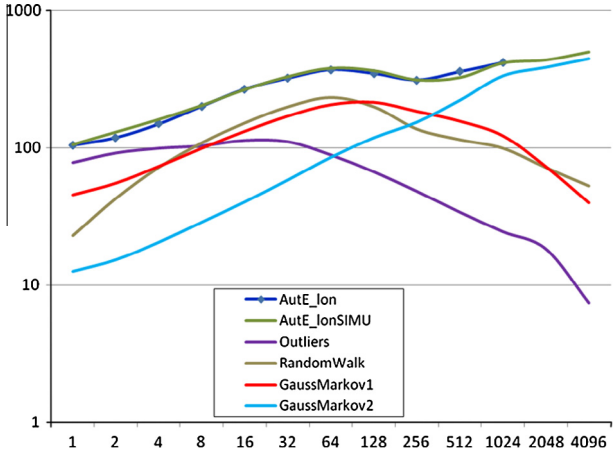


Fig. 4. Allan deviations for separately simulated noise terms for *AutE_lon*.

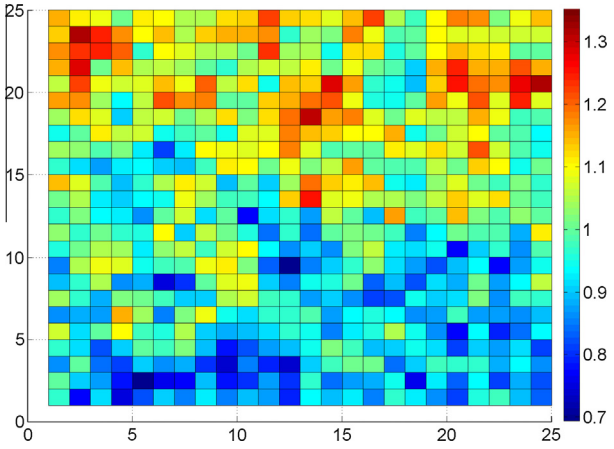


Fig. 6. Simulated error multiplier map named A-Map with spatial autocorrelation.

Table 5
Calculated Moran's I and Geary's C.

Test	Lag	Points	Neighb.	Moran's I	Geary's C
HP2006	20m	157	118	−0.013	1.039
VBS2006	20m	262	573	0.484	0.487
Aut, 2013	1m	14,125	186,685	0.829	0.310
HP, 2013	1m	18,008	395,883	0.031	0.974
A-Map	1.5	625	2352	0.610	0.381

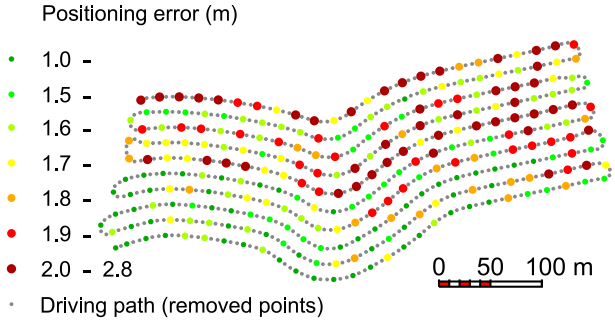


Fig. 5. Positioning error of selected points in VBS2006 test drive.

mean = 1, sill = 0.01 and range = 1 to construct the *A-Map*. The calculated map was multiplied by 0.888 to adjust the mean value of the random map to be 1. The Geary's C and Moran's I values calculated from the *A-Map* (0.61 and 0.31) were within the values determined from real data so the *A-Map* fulfilled our requirements. The standard deviation of the *A-Map* was 0.13.

In the simulator, we calculated an offset and scaled the *A-Map* to always cover the minimum bounding rectangle (MBR) of the input data. In practice, we scaled the *A-Map* to correspond to the size of the operational field, being able to simulate the effect of field surroundings. We also left a possibility to separately calculate different size of *A-Map* grids. The unmodified size of the *A-Map* is 25 hectares and the grid width is 20 × 20 m.

2.6. Structure of the simulator

We constructed the error simulator in Matlab Simulink environment. The simulator consists of five main parts. The generalized block design for latitude is presented in Fig. 7. The generalized block design for longitude is identical.

First are the data input blocks, which include error free coordinates, setup parameters, defined noise components and the grip map. The coordinates can be generated by simulation, or they can be a previously recorded accurate (RTK) measurements or real

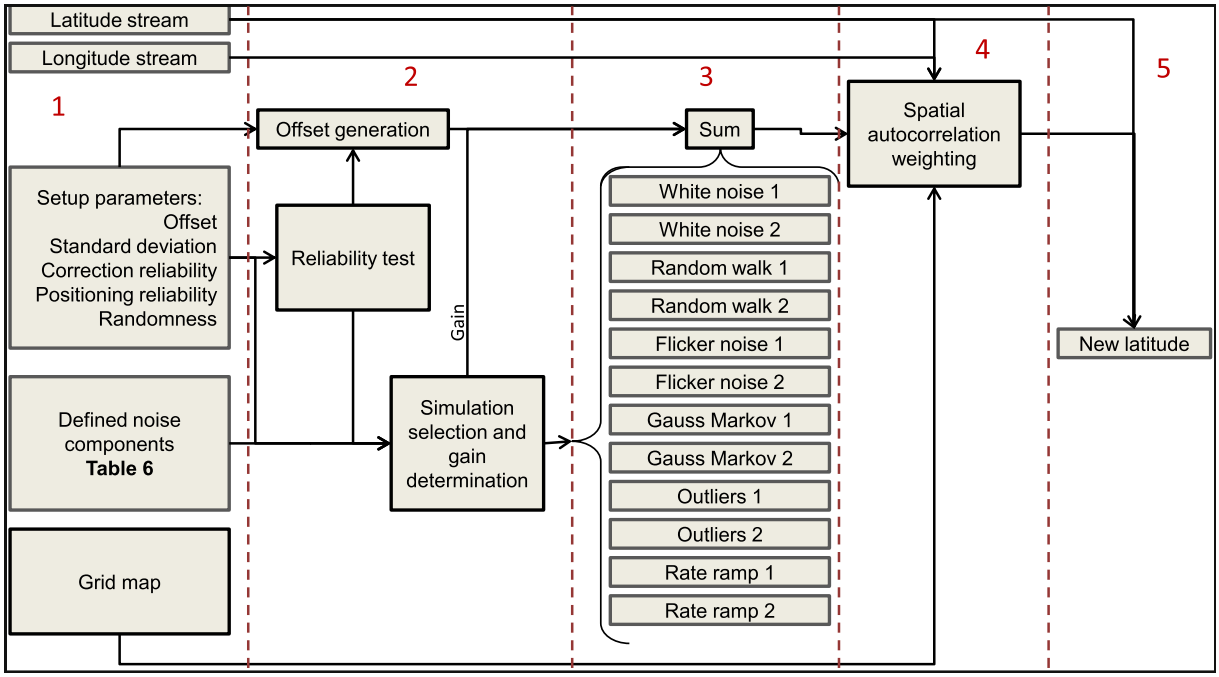


Fig. 7. Generalized block design of the constructed error simulator.

time measurements. The setup parameters define the requested level of accuracy and reliability so the simulations are not limited to equipment used in test measurements. The user can define the following parameters: offset, standard deviation, correction reliability, positioning reliability and randomness. The defined noise component's input is the data table of defined noise terms presented later in Table 6. The grid map input is the scaled *A-Map*.

The second block set selects the exact noise parameters for the simulation. The reliability test block determines if the simulation drops out to the poorer simulation: *RTK* to Float mode, *HP* to *Aut* mode. The duration of poorer mode was randomized to last from 2 to 100 s according to our Float-mode determinations, including a 0–1 m immediate leap in the position offset. The positioning reliability parameter determines if there will be a position output at all in some moments. The duration of that was defined to be between 0.1 and 20 s. The output of the reliability test block is the selected main simulation class. The simulation selection block selects the correct parameters from Table 6 based on the selected class. As the basis, the simulator selects the simulation with the closest standard deviation for both components: *Aut* 1.6 m, *HP*

Table 6
Noise components for all of the determined simulations.

Noise:		AutE_lon	AutE_lat	Aut_lon	Aut_lat	HP_lon	HP_lat	Float_lon	Float_lat	RTK_lon	RTK_lat
1	Update rate	3	1	1	1	1	1	1	1	1	1
	Update probability	0.5	1	1	1	1	1	1	1	1	1
	Reset time	100									
	Standard deviation	334	44	29	31	51	52	2.4	1.4	2	1.8
	Noise type	RW	WN	WN	FN	WN	WN	WN	WN	WN	WN
2	Update rate	59	150	640	10	2	13	4	4	5	9
	Update probability	1	1	1	0.5	1	0.1	0.5	0.5	0.5	0.1
	Reset time				250						
	Standard deviation	350	310	690	104	16	16	1.4	1.3	3	2.5
	Noise type	GM	GM	GM	RW	WN	FN	WN	WN	FN	FN
3	Update rate	2000	100	3200	63	19	440	6	6	35	150
	Update probability	1	1	1	0.5		1	1	1	0.25	1
	Reset time										
	Standard deviation	570	150	620	62	137	68	25	6	2.3	3.5
	Noise type	GM	GM	GM	WN	RW	GM	RW	RW	FN	FN
4	Update rate	5800	390		190	1020	1200	130	350	511	890
	Update probability	0.5	0.01		1	1	1	1	1	1	1
	Reset time	13									
	Standard deviation	350	195		110	190	55	−3	−3.5	7.9	7.5
	Noise type	OL	WN		GM	GM	GM	RR	RR	GM	GM
5	Update rate	2900	1000		950			500	500		
	Update probability	0.1	0.1		1			1	1		
	Reset time	10	10								
	Standard deviation	2900	100		115			8	1.3		
	Noise type	OL	OL		GM			RR	RR		

0.33 m, *RTK* 0.01 m. The block also calculates a gain based on the difference between requested standard deviation and the standard deviation of selected simulation class. If the positioning quality has dropped, the gain is set to 1. The offset generation block calculates a standard normal deviate out of the requested offset value being the same for the entire test. The possible position leap is separately added to the offset.

The third block set simulates the errors. The main block design of different noise types is presented in the following Fig. 8. First is an activation of a specific simulation, then the update rate and probability are set. The update probability is a probability of activating the simulation in time. Next the RW simulations could have a reset time for the drifting, which means that they could be stationary. Then the found standard deviations or weights are set. Then the actual noises are calculated followed by a random gain determined by the user (randomness). The randomness parameter adjusts the noise terms by randomizing their setup parameters with given and scaled threshold. Equal to the earlier parameter tuning cost function target for example, the 10% randomness will randomly calculate a 0–10% change to all of the current simulation parameters having randomizing effects on the total error.

Next, in the third block set the simulator sums up the separate simulations and multiplies the result of the selected noise to mach it to the requested standard deviation taking to account the found difference between coordinates. The offset value is then weighted and added separately to both coordinates. The sign for them is randomly selected.

The following fourth block set is the spatial autocorrelation weighting. This subsystem multiplies the amount of error according to the used *A-Map* in current coordinates. The effect of error is distributed between latitude and longitude by giving a Gaussian random weight between 0.8 and 1.2. Error for the latitude is multiplied by the weight and latitude is simultaneously divided by the weight. New random weight was calculated in every 30 s having effect only to longer cluster times. Because of the nominal 20-m grid size and about 10 km/h driving speed, the spatial autocorrelation increases noise in cluster times 4–8 and because of 30 s weight distribution variation, the white noise slightly increases in the cluster times between 256 and 512.

The last component of the error simulator is the erroneous coordinate output. The built-in delay is one measurement according to the simulation frequency. This is caused by the spatial autocorrelation determination.

3. Results

As the results of this study, we first present the defined base parameters and variables for the error generation. After that, we present the error simulation results.

3.1. Noise components

The following Table 6 presents the parameters for the noise components for the different types of positioning. The noise components for a single noise simulation are presented in one vertical line. The unit of update rate is 10 Hz and values are 10^{-8} degrees in WGS84 coordinates.

The simulated Allan standard deviation plots for all of the simulated noises are presented in Fig. 9 together with the original noises based on the test drives. The units of the time axis *X* is 0.1 s, and the unit of the *Y*-axis is 10^{-8} degrees in WGS84 coordinates to correspond to the operational values.

The following Table 7 presents a cluster time comparison between simulated and whole data set. Our simulation varied 6.4% in average compared to complete test drives. The cost function limit 10% was rather strict. We also compared the Allan variances of first and second half of each measured dataset, the standard deviation between them was 23.8% in average. Both halves in *HP* drive were identical and the standard deviations for coordinates were 11.9% and 18.7%.

The calculated overall PSD slopes are also presented in Table 7. The following Fig. 10 presents the PSD plots for each data. The presented moving average was calculated for 100 values. In these plots, the simulated data and measured GPS data should be as equal as possible. Differences in the overlapping curves would indicate differences in the noise power of certain frequencies. The high frequencies are on the right side of each plot. The differences in the minimum *x*-values are due to difference of the amount of points.

These PSD plots in overall showed that the moving average was a bit more constant in the PSD curves calculated from the simulated data. The difference between moving averages were 4% in average meaning that those curves were very similar. The greatest differences were in the *HP lat* plot where were peaks at $x \approx 1.2$ and $x \approx 2.2$ corresponding to about 3.5 Hz and 0.9 Hz frequencies. The corresponding Allan variance plot (Fig. 9) shows that our simulations were deviating in the same time regions. The mean distance and standard deviation between moving averages did not correlate with the Allan variance differences (Table 7).

Fig. 11 shows about eight minute part of the simulation of different noise components for the *AutE_lon* simulation. Also the sum of simulation components and the original *AutE_lon* measurement data are presented.

3.2. Simulator results

We tested the simulator by weakening the positioning data of a seeding work driven earlier with a RTK-GPS positioning system with successful 5 Hz data logging. We used the determined noise components of the Autonomous positioning (*Aut_lat/lon*) and the presented

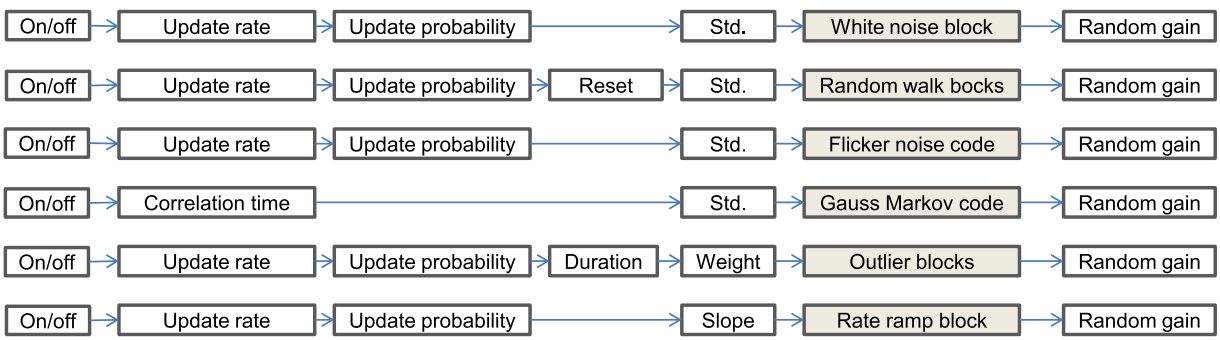


Fig. 8. Generalized block design for the detected Allan variance noise term construction.

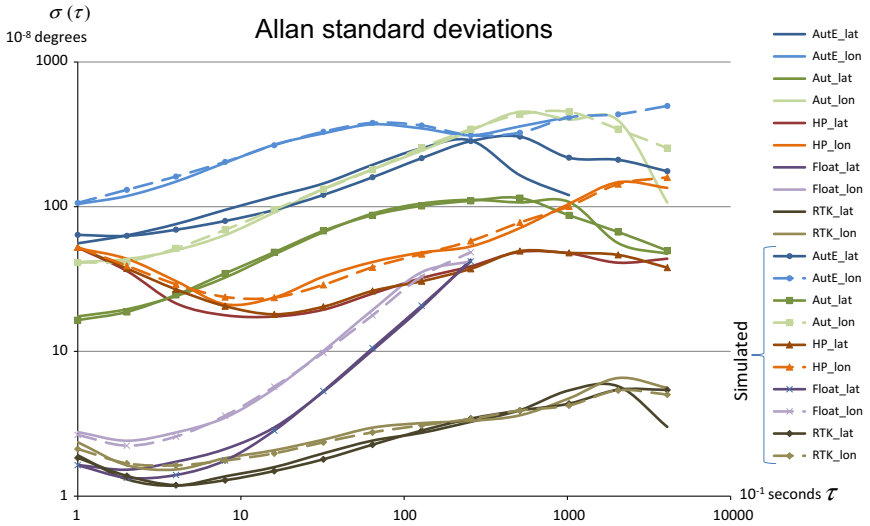


Fig. 9. Allan standard deviation plots of the test drives and simulation results.

Table 7
Comparison of difference between simulated and measured data.

	AutE_lon	AutE_lat	Aut_lon	Aut_lat	HP_lon	HP_lat	Float_lon	Float_lat	RTK_lon	RTK_lat	Mean	Measured 1/2 vs. 2/2
Mean error%	4.2	7.5	3.3	3.9	7.0	6.8	5.4	7.5	5.4	4.7	2.0	17.3
STD%	5.4	8.3	4.0	4.6	8.6	9.0	4.6	9.0	6.1	5.3	6.4	23.8
Simulated PSD slope	−1.6	−1.4	−1.9	−1.9	−1.1	−0.9	−2.1	−2.1	−1.5	−1.4		
Measured PSD slope	−1.7	−1.6	−1.8	−1.9	−1.2	−1.0	−1.8	−2.2	−1.6	−1.6		

A-Map. Fig. 12 shows a map of the seeding work with the illustrated total positioning error and the scaled *A-Map* grid as the background. The used *A-Map* caused 3.9% standard deviation in average for the Allan deviation plot from 1 to 128 cluster times. From 256 to 4096 cluster times, the deviation varied from −23% to +25%.

4. Discussion

This simulator was developed with agricultural tractors. The most case dependent factors were the driving speed (9 km/h), the level of environmental interference to the signals (neighboring forest), only two axes (no elevation) and the importance of spatial autocorrelation (constantly returning to a nearby area).

As the examination of Fig. 9 shows, we managed to construct a simulator which can simulate GNSS errors according to found Allan variance plots. The overall PSD slopes were also close to each other verifying the correct trend. As Figs. 11 and 12 (green[1] and blue curves) shows, constructed errors seem to model the actual error fairly well. However, there was a great difference between the Allan variances of first and second halves of the measurement data (17.3% on average). Also the difference between *Aut* and *AutE* was about 26%. This means that there is a great variation between different dynamic test drives, and the Allan variances are highly depending on the temporal positioning conditions during the tests. Our cost function limit for parameter optimization (less than 10%) was much smaller meaning that we are simulating the real test drives accurately. These figures are indicating that we can replicate our test drives, but tuning the variances and tolerances of simulator components to the universalisable level would require massive amount of test drives. However, the randomizing parameters in the simulating process made a great variation in total even without changing the input parameters (random seeds of the simulations, randomness in starting and resetting times, randomness in the error distribution between coordinates, the effect spatial autocorrelation map (4–25% std.). Especially the spatial autocorrelation affected really in the longer cluster times, where the Allan variance differences between test drives were much greater. This means that taking into account the spatial autocorrelation effect improves the realisticness of the simulation. In total, the difference between test drives and the difference and variation within simulations were close to each other meaning that the simulator is acting very realistically. Also the evaluation of PSD plots (4% difference in average) indicated that the simulator was acting well. However, some mark able variations in the PSD plots indicated that it would be beneficial to use the PSD evaluations already in the tuning process in the dynamic positioning environment.

The spatial autocorrelation tool worked as assumed. However we only manage to show how the spatial autocorrelation can be taken account. It would require more tests to determine a more generic way how the spatial autocorrelation affects the positioning and how it can be scaled. In our simulations, the spatial autocorrelation influenced equally the different noise components. In practice, the case might be different. The spatial autocorrelation value could also have had an effect on the reliability setup values by increasing the possibility of correction loss.

The usage of the simulator is rather straight forward. It is easy to change the parameters that change the simulation. Adding entirely new positioning simulation would require manual work with parameter tuning for finding the correct noise terms, but after that the simulator would easily be able to exploit it.

There are three questions left outside the scope of this research: (1) how much there is actual variation in noise components with a

[1] For interpretation of color in Figs. 11 and 12, the reader is referred to the web version of this article.

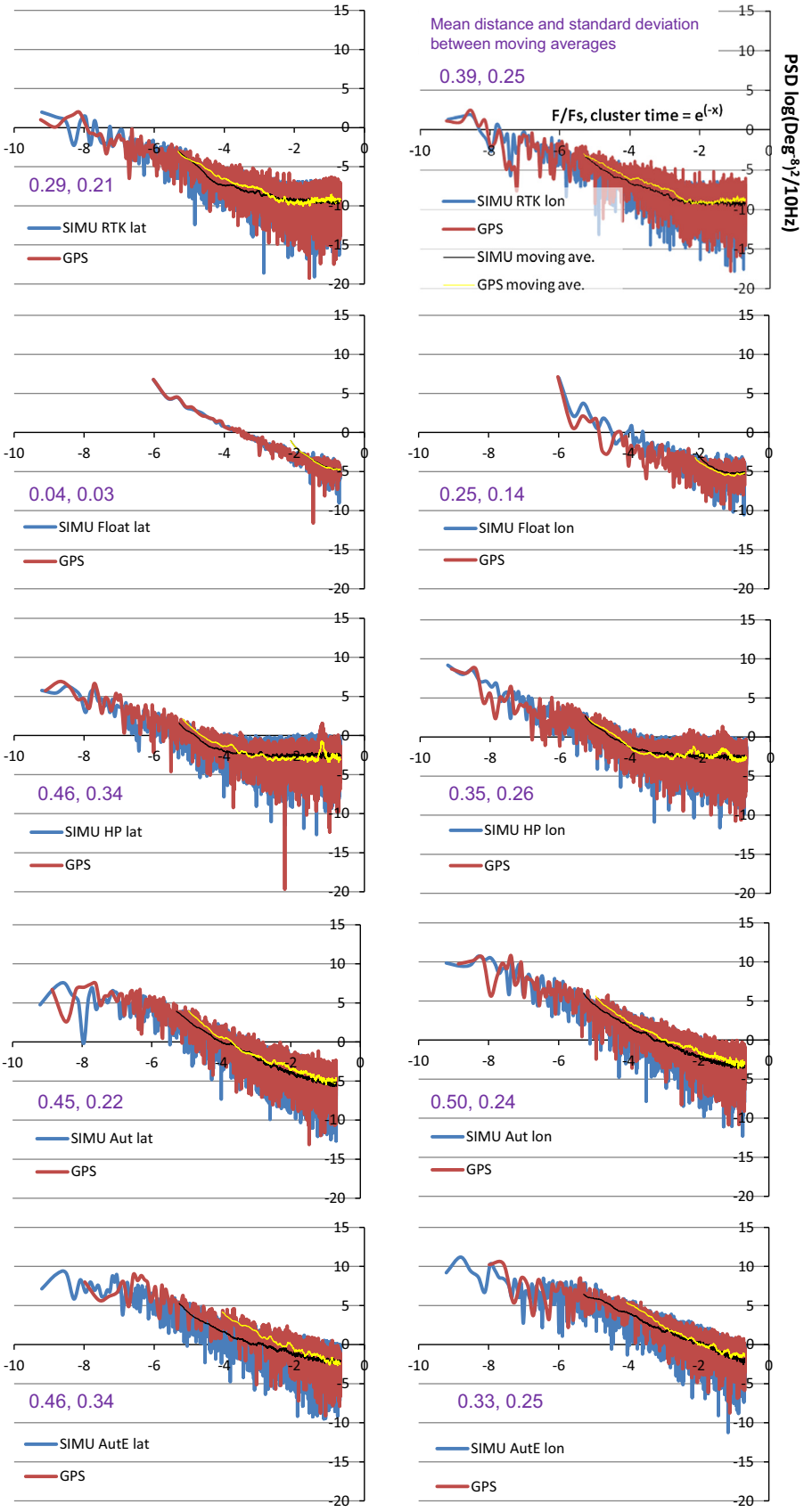


Fig. 10. Power spectral density plots of simulated and real test data.

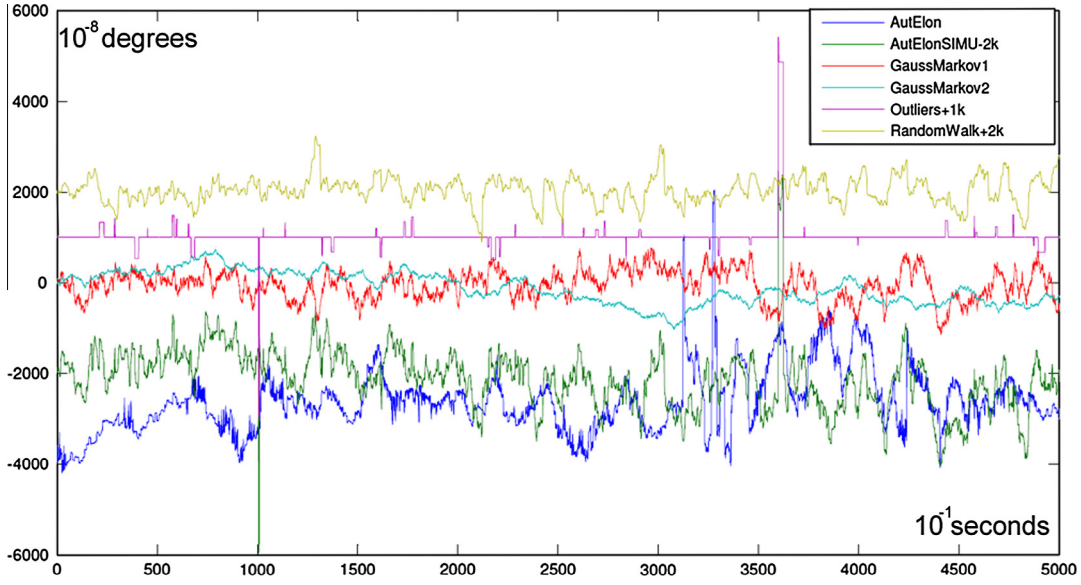


Fig. 11. Original data and all of the noise components for *AutE_lon* simulation separately and as aggregated.

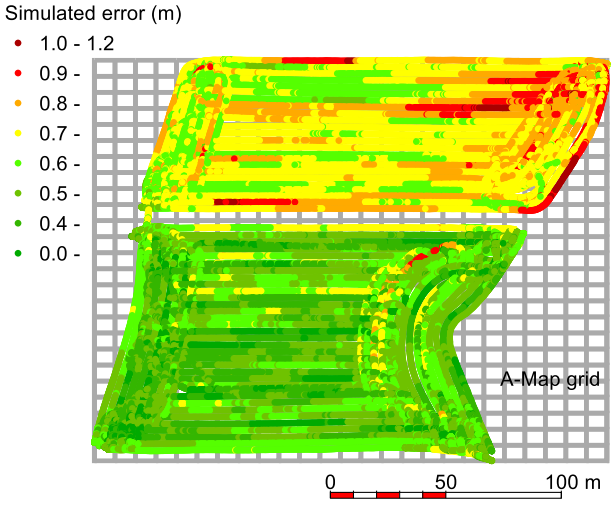


Fig. 12. A map presenting simulated error for an applied seeding work.

single GNSS setup in dynamic environment, (2) would the spatial autocorrelation act differently for different noise components and (3) what kind of effect is caused by the usage of RTK-GNSS positioning as a reference? Getting an answer to the first and second question would need more dynamic tests. Answering the third question would require a test infrastructure such as a solid rail track or laser networks. The last drawback of the presented research is that some of the used test data were rather old. The GNSS positioning has been developing all the time. However more resent data would not have an effect on the simulator design and test measurements will always reflect only the past conditions.

5. Conclusions

We constructed a realistic GNSS error simulator for mobile machines, including local differences in the amount of error. As such, it can flexibly support different sensor fusion in the steering and navigation automation developments. We also confirmed that Allan variance is a feasible way to determine the structure of the errors, and the errors can be reconstructed with it. We replicated the found Allan variances by combining different noise terms: Gaussian White noise, Gaussian Random walk, Flicker noise, the First order Gauss–Markov process, Rate ramp and outliers. However, it could be beneficial to examine PSD curves in the parameter optimization phase.

The constructed simulator is most suitable for navigation and spatial application development purposes with the agricultural machinery and with other mobile machinery working on a rather open view. This simulator could be easily used also for mobile machinery working on a rather open view (e.g. snow ploughing, grass cutting, bulldozer works, square cleaning) and could have some adaptability for vehicles such as forest machinery, road maintenance and military vehicles. Expansion to the third dimension would be rather straight forward. In agricultural field works, the error simulations can also be applied for position determination with variable rate applications (VRA), with other source data for machinery output adjustments and with position reporting (Mousazadeh, 2013 and Kaivosoja et al., 2014). In those cases, simulations could reveal how successful the work would have been with different positioning accuracies.

There are several ways how to improve the simulator but at this form, it will fulfill the need of the error simulator that will support the sensor fusion development and system requirement determinations. This work also presents in high detail that how does the GNSS positioning behave within farming operations.

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